Low-level EEG image reconstruction: THINGS vs. our lab data

Target: the coarse 8x8 colour layout of the image (no detail, no semantics). Linear ridge readouts. 8 examples per dataset, drawn at random.
Score = coarse8: correlation of the reconstructed 8x8 layout with the true one, relative to the other images of the set (0 = chance, 1 = perfect).

THINGS-EEG2

public benchmark

- Research-grade amplifier, 31 StarStim channels used
- 10 people, rapid series of 5 images per second
- Readout trained on 16,448 images
- Each test image: 80 repeats x 10 people

coarse8 +0.440

800 trials / image

Old lab data

May-June 2026

- StarStim 32 consumer cap, 31 channels
- 6 people, single images, ~4 s each
- Readout trained on these data (8-fold image cross-validation)
- ~1.25 trials per image per person, pooled over 6 people

coarse8 +0.079

~7 trials / image

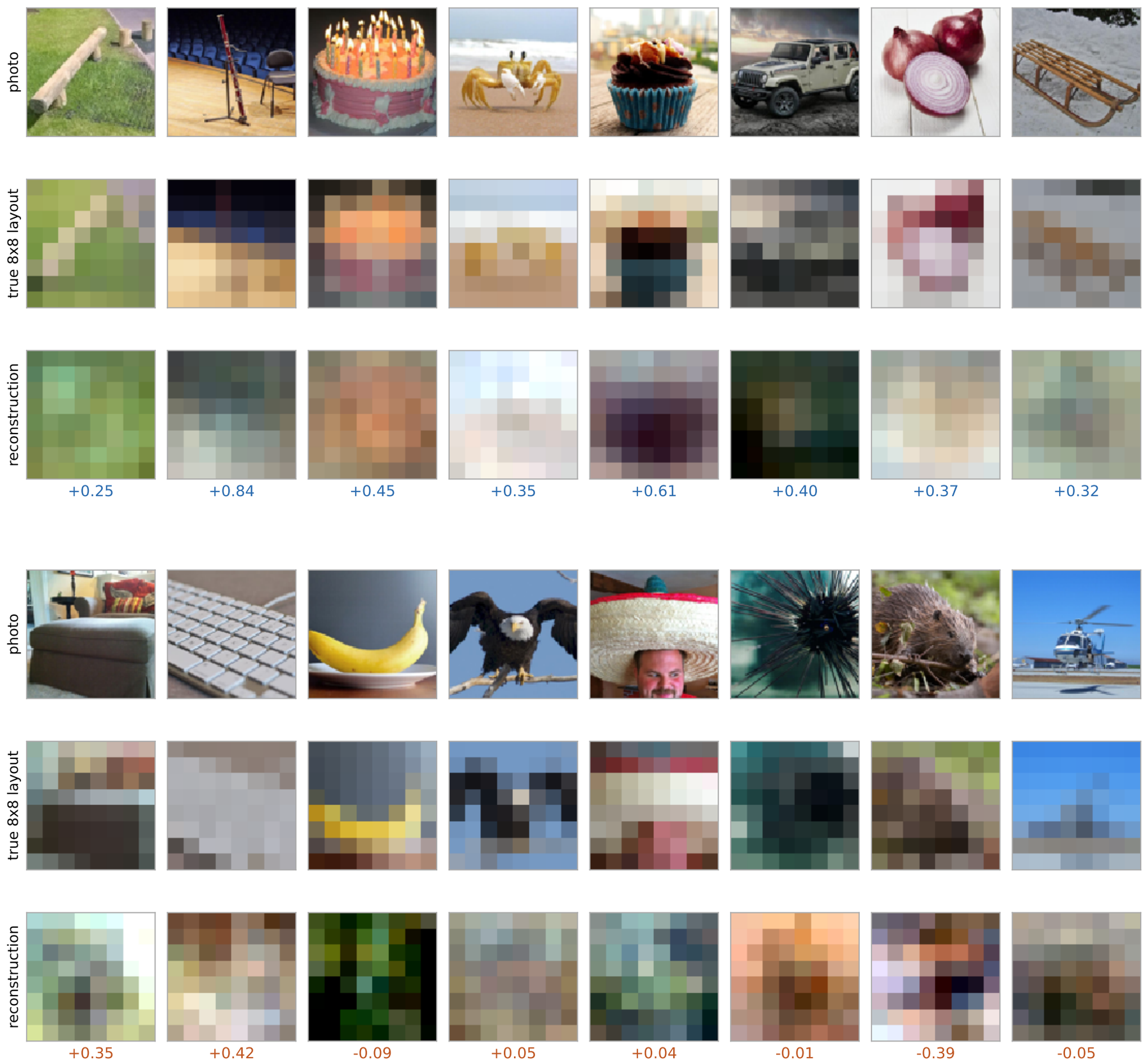
New lab sessions

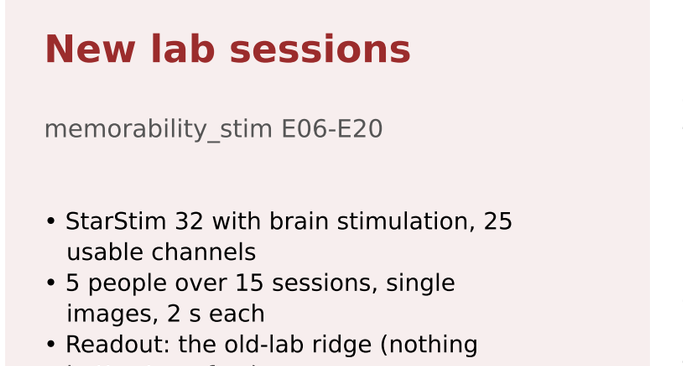
memorability_stim E06-E20

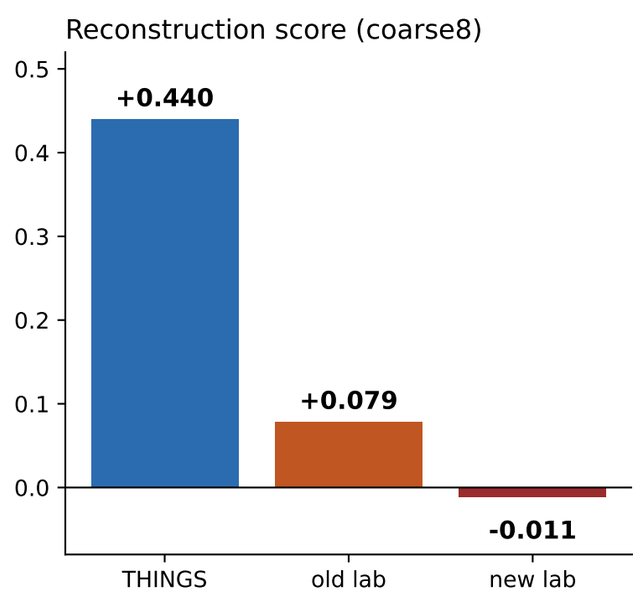
- StarStim 32 with brain stimulation, 25 usable channels
- 5 people over 15 sessions, single images, 2 s each
- Readout: the old-lab ridge (nothing better transfers)
- Pulse-free blocks only

coarse8 -0.011

2-4 trials / image







Why the lab reconstructions fail

1 Too few repeats per image

Single trials are at chance in every lab dataset; the signal only appears after averaging: 800 trials per THINGS test image vs ~7 (old lab) and 2-4 (new sessions).

2 Too little data per person

Decoding scales with trials per person (~5,000 needed, Banville et al. 2025). Lab people give ~100-200, and the readout must work on people it never saw.

3 THINGS does not transfer

Rapid 5-per-second series vs single images, research amplifier vs consumer cap, other people: the THINGS network, its features, a fine-tuned adapter and a THINGS-weight prior all fail on lab data.

4 Stimulation costs signal

Six stimulation electrodes are excluded (PO3/PO4 among them), P8/O2/FC2/Cz go flat in pulse runs, and pulse artifacts leave only the pulse-free blocks.

What would fix it

- A dedicated block before any stimulation: ~40 shared images x 30-40 repeats, shown rapidly (~200 ms image + 200-400 ms blank), about 15 minutes.
- The same few people across sessions, so each person accumulates data.
- Linear readouts per person, averaged as predictions; target the coarse layout (top-bottom lightness is the most decodable property).

Reconstructions are contrast-enhanced for display (one gain per dataset). THINGS ridge penalty chosen on its test images (1 of 10 values).
Old lab: 8-fold image cross-validation. New sessions: readout trained on the old lab data, scored within each session.